UNIT 2 MINI PROJECT

Enterprise Student Record and Academic Management Platform

AGILE / SCRUM PROJECT PLANNING

B.Tech Mini-Project Documentation

Focus: Product Vision • Scrum Planning • Backlog • Sprints • Agile Metrics • Release Roadmap

Prepared as a project-planning deliverable

Document Overview

Visual design approach: figures use restrained color accents to distinguish workflow stages, status, priority, and outcomes while keeping the document suitable for formal academic submission.

This Unit 2 document translates the Enterprise Student Record and Academic Management Platform into an Agile delivery plan. It focuses on product vision, value-driven backlog design, Scrum responsibilities, estimation, sprint execution, inspection and adaptation, framework selection, metrics, risks, and release planning. It intentionally avoids repeating the detailed system-analysis material that belongs in Unit 1.

Part	Primary purpose
1–3	Agile context, business problem and product vision
4–6	Scrum team, roles, epics and user stories
7–9	Acceptance criteria, backlog and estimation
10–12	Sprint planning, sprint backlog, Scrum board and burndown
13–16	Ceremonies, Definition of Done, framework comparison and metrics
17–20	Roadmap, risks, traceability, final summary and checklists

1. Introduction

Agile software development is an iterative approach in which a product is built in small, valuable increments rather than being treated as one large delivery at the end. Agile emphasizes collaboration, working software, frequent feedback, transparency and the ability to adapt when requirements become clearer.

For an academic management platform, these characteristics are particularly useful. Different users—students, faculty, academic administrators and examination staff—interact with the product in different ways. A requirement that appears correct during initial planning may need adjustment after users see an actual workflow. Incremental development allows the team to validate the highest-value workflows early and reduce the risk of spending large amounts of effort on features that later need redesign.

Traditional Sequential Development vs Agile

Dimension	Sequential approach	Agile approach
Requirements	Defined heavily at the beginning	Refined continuously as learning occurs
Delivery	Large release near the end	Frequent increments
Feedback	Often concentrated at milestones	Built into each sprint/review cycle
Change	Can be expensive late in the project	Expected and managed through backlog reprioritization
Testing	May become concentrated late	Testing is part of each increment
Risk	Problems can remain hidden for longer	Risks become visible through short cycles
Customer/user value	May arrive late	Highest-value capabilities delivered earlier

Why Agile Fits This Project

- Academic workflows contain business rules that benefit from early validation with representative users.
- Authentication, attendance, marks and results have different dependencies, making incremental sequencing practical.
- A small student development team can focus on a limited sprint goal instead of attempting the complete platform at once.
- Frequent reviews provide opportunities to refine usability and clarify ambiguous academic rules.
- Agile metrics provide evidence about progress without assuming that every original estimate will remain correct.

2. Business Problem and Product Context

The platform addresses the fragmentation and manual effort that can occur when student information, course registration, attendance, marks and academic reporting are handled through disconnected processes. The purpose of this section is not to repeat Unit 1, but to identify the problem characteristics that influence Agile planning.

Problem	Users affected	Agile implication
Repeated/manual student-record updates	Students, administrators	Prioritize a single reliable student record
Course information and enrollment coordination	Students, faculty, administrators	Deliver course workflows after core identity and access
Attendance recording and visibility	Faculty, students, administrators	Validate the workflow with users early
Marks and result calculation	Faculty, examination cell, students	Treat calculation rules as acceptance criteria
Delayed notifications	Students, faculty	Add after core academic data flows are stable
Reporting effort	Administrators, department heads	Build reports from validated underlying data

Why Requirements May Change

- Users may discover that a workflow requires fewer steps or different permissions after seeing a prototype.
- Academic rules can contain exceptions that are difficult to capture completely in an initial discussion.
- Reporting needs may become clearer once attendance and marks data are available.
- Technical constraints discovered during development may require a story to be split or reordered.
- Security and access-control feedback may change which roles can perform particular actions.

Agile Delivery Principle for This Project

The team therefore plans around business outcomes and thin vertical slices of functionality. Each sprint should produce an increment that can be demonstrated, tested and evaluated—not simply a collection of unfinished technical tasks.

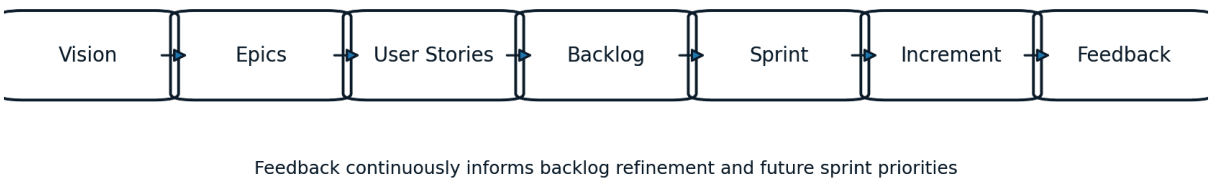


Figure 1. Agile product development flow for the academic management platform.

3. Agile Product Vision

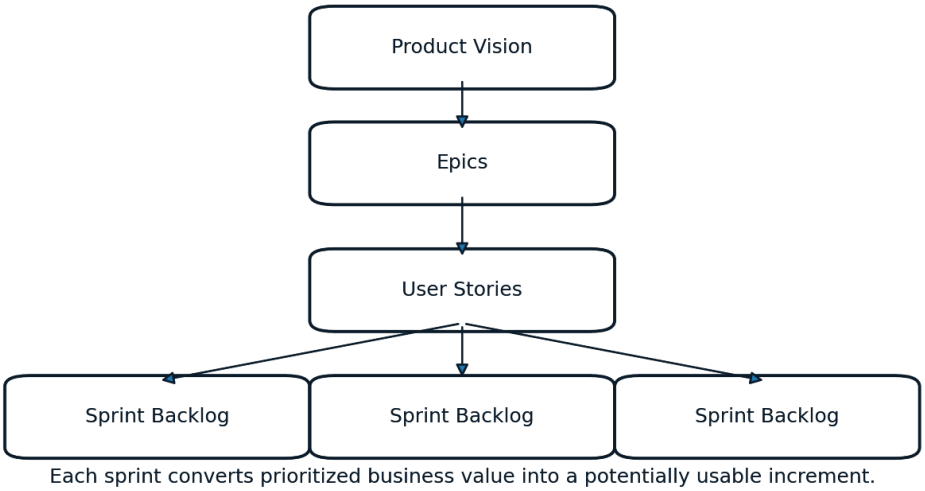


Figure 2. Product vision translated into epics, stories, sprint backlogs and increments.

Product Objective

Create a secure, role-aware and usable web-based platform that centralizes the most important student academic workflows and delivers measurable improvements in data accessibility, workflow consistency and reporting efficiency.

Primary Users and Outcomes

User / stakeholder	Desired outcome
Students	View accurate profiles, courses, attendance and academic results without relying on disconnected records.
Faculty	Record attendance and marks efficiently and see only the academic data relevant to assigned responsibilities.
Academic Administrators	Maintain student/course information and produce consistent academic reports.
Examination Cell	Manage marks/result workflows with traceable calculations and controlled access.
IT Support	Operate a maintainable system with clear roles, logs and manageable deployment practices.
Department Heads / Management	Receive useful summaries for academic monitoring and decision support.

Measurable Success Criteria

- Core users can authenticate and reach only the functions permitted by their roles.
- A faculty member can complete a representative attendance entry without duplicate records for the same course/date/session.

- Marks entered for an authorized course can be validated and reflected in the corresponding academic result workflow.
- Students can access their permitted academic information through a consistent interface.
- Priority MVP stories meet their acceptance criteria and Definition of Done before release.
- Stakeholder feedback from sprint reviews is captured as backlog changes rather than informal side requests.

Product Vision Statement

“For higher-education users who need dependable academic information, the Enterprise Student Record and Academic Management Platform will provide a secure and progressively delivered workspace for student records, course enrollment, attendance, academic performance, results, notifications and reporting, so that essential academic workflows become easier to perform, verify and monitor.”

Constraints and Assumptions

- Scope is suitable for a B.Tech mini-project and therefore emphasizes an MVP rather than institution-wide enterprise scale.
- Team capacity is limited and multiple technical responsibilities may be shared.
- Academic rules such as grading formulas are treated as configurable project assumptions and must be confirmed before implementation.
- No claim is made that the institution already uses this proposed platform; it is a project design/planning artifact.

4. Scrum Team Structure

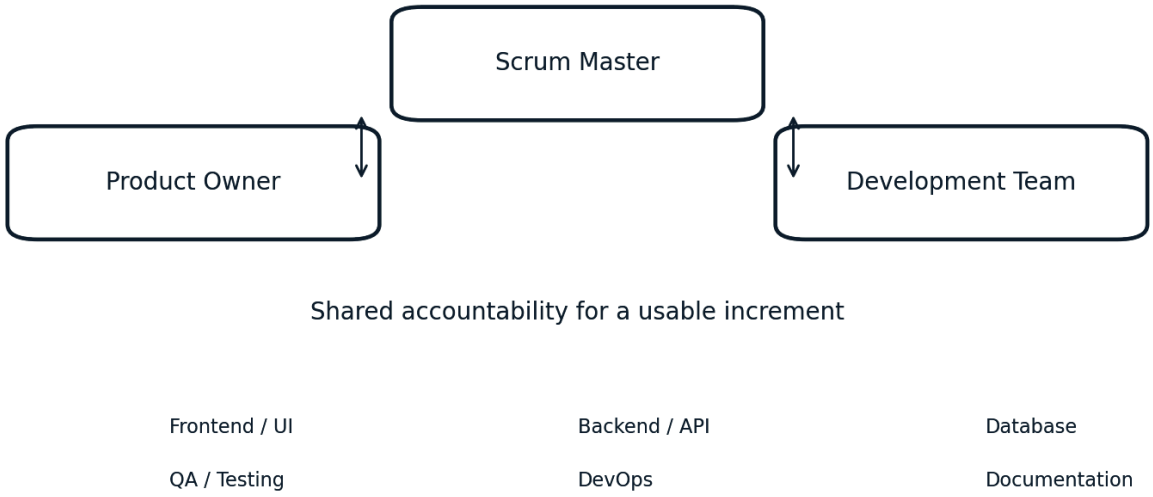


Figure 3. Scrum Team structure and shared development responsibilities.

The Scrum Team is intentionally small enough for a college project while still covering the skills required to produce a usable increment. In a small team, the labels below represent responsibilities rather than six separate full-time people.

Role / responsibility area	Primary responsibilities	Typical project contribution
Product Owner	Vision, stakeholder communication, backlog ownership, prioritization, acceptance	Defines value, orders stories, clarifies requirements and accepts completed work.
Scrum Master	Scrum process, facilitation, impediment removal, team collaboration	Facilitates ceremonies, protects focus and makes process problems visible.
Frontend / UI	Screens, forms, navigation, client validation, usability	Builds student/faculty/admin-facing workflows.
Backend / API	Business logic, APIs, authorization, validation	Implements academic workflows and service interfaces.
Database	Schema, relationships, constraints, queries, data integrity	Maintains reliable student/course/attendance/marks data.
QA / Testing	Test cases, acceptance checks, regression, defect reporting	Verifies stories against acceptance criteria and DoD.
DevOps / Deployment	Repository, CI, environments, builds, deployment	Keeps the project reproducible and deployable.
Documentation	Story detail, decisions, test evidence, release notes	Maintains project knowledge and traceability.

Scrum Team Responsibility Matrix

Activity	PO	SM	Development / QA
Product vision	A/R	C	C
Backlog ordering	A/R	C	C
Story refinement	R	F	R
Estimation	C	F	R
Sprint planning	R	F	R
Daily Scrum	C	F	R
Implementation	C	C	R
Acceptance verification	A/R	C	R
Sprint review	A/R	F	R
Retrospective	C	F	R

Legend: A = accountable, R = responsible, C = consulted, F = facilitates. In a small student team, one person may hold more than one practical responsibility, but accountability for the Product Owner and Scrum Master accountabilities should remain clear.

5. Project Epics

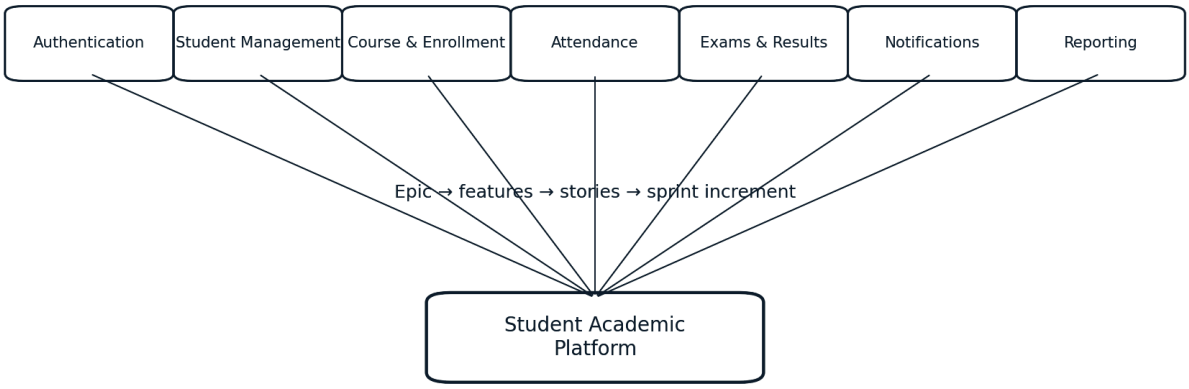


Figure 4. Epic hierarchy for the Enterprise Student Record and Academic Management Platform.

ID	Epic	Business purpose	User value	Major features	Dependencies	Priority	Target
EP-01	Authentication & User Management	Secure identity and role-based access	Users can enter the system safely and access only permitted functions	Registration, login, logout, password reset, RBAC	Foundation for protected modules	Must Have	MVP / Release 1
EP-02	Student Management	Maintain a reliable student record	Users can find and maintain relevant student information	Add/view/search/update student, academic information	Depends on authentication	Must Have	MVP / Release 1
EP-03	Course & Enrollment	Coordinate courses and student enrollment	Students and staff can manage course participation	Course CRUD, details, allocation, enrollment	Depends on student identity and roles	Must Have	MVP / Release 1
EP-04	Attendance Management	Record and monitor attendance	Faculty can record attendance and students can monitor status	Attendance entry, view, percentage, reports, shortage alerts	Depends on courses and enrollment	Must Have	Release 2
EP-05	Examination & Performance	Manage marks and academic outcomes	Authorized staff can enter marks and produce consistent results	Marks, grades, SGPA/CGPA, result sheets	Depends on students/courses	Must Have	Release 2

ID	Epic	Business purpose	User value	Major features	Dependencies	Priority	Target
EP-06	Notifications	Provide timely academic alerts	Users receive relevant attendance, exam and result notifications	Attendance alerts, exam notifications, result notifications	Depends on source events	Should Have	Release 3
EP-07	Reporting & Analytics	Turn academic data into useful summaries	Staff can inspect attendance, performance and course information	Student, attendance, performance and course reports	Depends on validated data	Should Have	Release 3

6. User Story Repository

The repository below is intentionally broad enough to represent the major product workflows. Stories are written as user outcomes rather than technical tasks. Story points are relative estimates and are refined during planning.

ID	Epic	Role	User story	Business value	Priority	SP	Dependency
US-AUT-01	EP-01	Student/Staff	As a user, I want to register an account using approved details, so that I can access the platform.	Identity access	Must	5	—
US-AUT-02	EP-01	All users	As a user, I want to log in securely, so that I can access my authorized academic functions.	Secure access	Must	3	US-AUT-01
US-AUT-03	EP-01	All users	As a user, I want to log out, so that my session is closed when I finish.	Security	Must	1	US-AUT-02
US-AUT-04	EP-01	Admin	As an administrator, I want to assign roles, so that access matches responsibilities.	Access control	Must	5	US-AUT-02
US-AUT-05	EP-01	User	As a user, I want to reset a forgotten password, so that I can recover access.	Continuity	Should	5	US-AUT-02
US-STU-01	EP-02	Admin	As an administrator, I want to add a student record, so that the academic profile exists in the system.	Data creation	Must	5	US-AUT-04
US-STU-02	EP-02	Authorized staff	As an authorized user, I want to search for a student, so that I can quickly	Efficiency	Must	3	US-AUT-04

ID	Epic	Role	User story	Business value	Priority	SP	Dependency
			locate the correct record.				
US-STU-03	EP-02	Student	As a student, I want to view my profile, so that I can verify my academic information.	Transparency	Must	3	US-AUT-02
US-STU-04	EP-02	Admin	As an administrator, I want to update permitted student details, so that records remain current.	Data quality	Must	5	US-STU-01
US-CRS-01	EP-03	Admin	As an administrator, I want to create and update courses, so that the course catalogue remains accurate.	Course management	Must	5	US-AUT-04
US-CRS-02	EP-03	Faculty	As a faculty member, I want to see assigned courses, so that I work only with relevant classes.	Usability	Must	3	US-CRS-01
US-CRS-03	EP-03	Student	As a student, I want to view available courses, so that I can choose appropriate enrollment options.	Choice	Must	3	US-CRS-01
US-CRS-04	EP-03	Student	As a student, I want to enroll in an available course, so that my academic schedule is recorded.	Enrollment	Must	5	US-CRS-03
US-ATT-01	EP-04	Faculty	As a faculty member, I want to record attendance for an assigned course and date, so that attendance	Core workflow	Must	8	US-CRS-02, US-CRS-04

ID	Epic	Role	User story	Business value	Priority	SP	Dependency
			remains accurate.				
US-ATT-02	EP-04	Student	As a student, I want to view my attendance percentage, so that I can monitor attendance status.	Awareness	Must	3	US-ATT-01
US-ATT-03	EP-04	Admin	As an administrator, I want an attendance report, so that shortage cases can be monitored.	Monitoring	Should	5	US-ATT-01
US-ATT-04	EP-04	System/Admin	As a system user, I want shortage alerts based on a configured threshold, so that attendance issues are noticed early.	Early action	Should	5	US-ATT-02
US-EXM-01	EP-05	Faculty	As a faculty member, I want to enter marks for an assigned course, so that academic performance is recorded.	Core workflow	Must	8	US-CRS-02
US-EXM-02	EP-05	Faculty/Admin	As an authorized user, I want marks to be validated before saving, so that invalid academic data is reduced.	Data quality	Must	5	US-EXM-01
US-EXM-03	EP-05	System	As the system, I want to calculate grades from configured rules, so that results are consistent.	Consistency	Must	8	US-EXM-02
US-EXM-04	EP-05	Student	As a student, I want to view my result, so that I	Transparency	Must	5	US-EXM-03

ID	Epic	Role	User story	Business value	Priority	SP	Dependency
			can understand my academic outcome.				
US-EXM-05	EP-05	Student/Admin	As a user, I want SGPA/CGPA to be calculated from configured inputs, so that summary performance is available.	Summary	Should	8	US-EXM-03
US-NOT-01	EP-06	Student	As a student, I want attendance shortage notifications, so that I can respond before the issue becomes serious.	Timeliness	Should	5	US-ATT-04
US-NOT-02	EP-06	Student	As a student, I want examination notifications, so that I know important academic events.	Timeliness	Could	3	US-EXM-04
US-NOT-03	EP-06	Student	As a student, I want result notifications, so that I know when results are available.	Timeliness	Could	3	US-EXM-04
US-REP-01	EP-07	Admin	As an administrator, I want student reports, so that academic records can be reviewed systematically.	Reporting	Should	5	US-STU-04
US-REP-02	EP-07	Admin	As an administrator, I want attendance reports, so that trends and shortage cases can be reviewed.	Reporting	Should	5	US-ATT-03
US-REP-03	EP-07	Department Head	As a department	Decision support	Should	8	US-EXM-05

ID	Epic	Role	User story	Business value	Priority	SP	Dependency
			head, I want performance summaries, so that academic performance can be monitored.				
US-REP-04	EP-07	Admin	As an administrator, I want course reports, so that enrollment and course information can be reviewed.	Reporting	Could	5	US-CRS-04

7. Acceptance Criteria

Acceptance criteria make a story testable and reduce ambiguity. The examples below cover the highest-value workflows. Criteria are written so a reviewer can determine whether the story is complete without relying on vague statements such as “works properly.”

Story	Acceptance focus	Measurable criteria
US-AUT-02	Secure Login	Given a registered active user, when valid credentials are submitted, then an authenticated session is created. Invalid credentials must not create a session; the user receives a clear error; protected pages require authentication.
US-AUT-04	Role Assignment	Given an administrator with role-management permission, when a valid role is assigned to a user, then the role is stored and the next authorization check reflects the new permission. Unauthorized users cannot change roles.
US-STU-01	Add Student	Given an authorized administrator, when valid mandatory student information is submitted, then a unique student record is created. Duplicate identifiers are rejected and validation errors are displayed clearly.
US-CRS-04	Course Enrollment	Given an eligible student and an available course, when the student confirms enrollment, then the enrollment is stored once. An already-enrolled student cannot create a duplicate enrollment.
US-ATT-01	Record Attendance	Given an authenticated faculty member assigned to the selected course, when attendance is submitted for a valid date/session, then the record is saved. Duplicate course/date/session entries are prevented and the user receives confirmation.
US-EXM-02	Validate Marks	Given an authorized marks editor, when marks are entered, then values outside configured limits are rejected and valid values are saved only after validation succeeds.
US-EXM-03	Calculate Grade	Given validated marks and a configured grading rule, when the result is calculated, then the grade is produced consistently with that rule and the calculation can be traced to the source marks.
US-EXM-04	View Result	Given a student with a published result, when the student opens the result view, then only that student's authorized result is displayed.
US-ATT-04	Shortage Alert	Given an attendance threshold configured by the project rules, when a student's percentage falls below the threshold, then

Story	Acceptance focus	Measurable criteria
		the shortage status is identified and the notification workflow can generate an alert.

Definition of Done

- Story implementation is complete and integrated into the project branch/repository.
- Code has been reviewed by another team member where practical.
- Relevant unit and integration tests pass; acceptance scenarios have been checked.
- Authorization and validation rules relevant to the story have been tested.
- UI behavior is usable and validation/error states are understandable.
- Documentation or configuration affected by the story is updated.
- No known critical defect remains open against the story.
- Product Owner has verified the acceptance criteria and accepts the increment.



Figure 5. User-story workflow from need identification through refinement and acceptance.

8. Prioritized Product Backlog

The backlog is ordered using business value, dependency, user impact, risk and effort. Must-Have stories form the MVP foundation; Should-Have stories add important operational value after the core workflow is stable; Could-Have items are deferred when capacity is limited.

Priority	Story ID	Epic	User Story	Business value	SP	Dependency	Sprint	Status
Must	US-AUT-01	EP-01	As a user, I want to register an account using approved details, so that I can access the platform.	Identity access	5	—	Sprint 1	Planned
Must	US-AUT-02	EP-01	As a user, I want to log in securely, so that I can access my authorized academic functions.	Secure access	3	US-AUT-01	Sprint 1	Planned
Must	US-AUT-03	EP-01	As a user, I want to log out, so that my session is closed when I finish.	Security	1	US-AUT-02	Sprint 1	Planned
Must	US-AUT-04	EP-01	As an administrator, I want to assign roles, so that access matches responsibilities.	Access control	5	US-AUT-02	Sprint 1	Planned
Should	US-AUT-05	EP-01	As a user, I want to reset a forgotten password, so that I can recover access.	Continuity	5	US-AUT-02	Sprint 1	Planned
Must	US-STU-01	EP-02	As an administrator, I want to add a student record, so that the academic profile exists in the system.	Data creation	5	US-AUT-04	Sprint 1	Planned
Must	US-STU-02	EP-02	As an authorized user, I want to search for a student, so that I can quickly	Efficiency	3	US-AUT-04	Sprint 1	Planned

Priority	Story ID	Epic	User Story	Business value	SP	Dependency	Sprint	Status
			locate the correct record.					
Must	US-STU-03	EP-02	As a student, I want to view my profile, so that I can verify my academic information.	Transparency	3	US-AUT-02	Sprint 1	Planned
Must	US-STU-04	EP-02	As an administrator, I want to update permitted student details, so that records remain current.	Data quality	5	US-STU-01	Sprint 1	Planned
Must	US-CRS-01	EP-03	As an administrator, I want to create and update courses, so that the course catalogue remains accurate.	Course management	5	US-AUT-04	Sprint 2	Planned
Must	US-CRS-02	EP-03	As a faculty member, I want to see assigned courses, so that I work only with relevant classes.	Usability	3	US-CRS-01	Sprint 2	Planned
Must	US-CRS-03	EP-03	As a student, I want to view available courses, so that I can choose appropriate enrollment options.	Choice	3	US-CRS-01	Sprint 2	Planned
Must	US-CRS-04	EP-03	As a student, I want to enroll in an available course, so that my academic schedule is recorded.	Enrollment	5	US-CRS-03	Sprint 2	Planned
Must	US-ATT-01	EP-04	As a faculty member, I want to record attendance for an assigned course and date, so that attendance	Core workflow	8	US-CRS-02, US-CRS-04	Sprint 3	Planned

Priority	Story ID	Epic	User Story	Business value	SP	Dependency	Sprint	Status
			remains accurate.					
Must	US-ATT-02	EP-04	As a student, I want to view my attendance percentage, so that I can monitor attendance status.	Awareness	3	US-ATT-01	Sprint 3	Planned
Should	US-ATT-03	EP-04	As an administrator, I want an attendance report, so that shortage cases can be monitored.	Monitoring	5	US-ATT-01	Sprint 3	Planned
Should	US-ATT-04	EP-04	As a system user, I want shortage alerts based on a configured threshold, so that attendance issues are noticed early.	Early action	5	US-ATT-02	Sprint 3	Planned
Must	US-EXM-01	EP-05	As a faculty member, I want to enter marks for an assigned course, so that academic performance is recorded.	Core workflow	8	US-CRS-02	Sprint 4	Planned
Must	US-EXM-02	EP-05	As an authorized user, I want marks to be validated before saving, so that invalid academic data is reduced.	Data quality	5	US-EXM-01	Sprint 4	Planned
Must	US-EXM-03	EP-05	As the system, I want to calculate grades from configured rules, so that results are consistent.	Consistency	8	US-EXM-02	Sprint 4	Planned
Must	US-EXM-04	EP-05	As a student, I want to view my result, so that I	Transparency	5	US-EXM-03	Sprint 4	Planned

Priority	Story ID	Epic	User Story	Business value	SP	Dependency	Sprint	Status
			can understand my academic outcome.					
Should	US-EXM-05	EP-05	As a user, I want SGPA/CGPA to be calculated from configured inputs, so that summary performance is available.	Summary	8	US-EXM-03	Sprint 4	Planned
Should	US-NOT-01	EP-06	As a student, I want attendance shortage notifications, so that I can respond before the issue becomes serious.	Timeliness	5	US-ATT-04	Sprint 5	Planned
Could	US-NOT-02	EP-06	As a student, I want examination notifications, so that I know important academic events.	Timeliness	3	US-EXM-04	Sprint 5	Planned
Could	US-NOT-03	EP-06	As a student, I want result notifications, so that I know when results are available.	Timeliness	3	US-EXM-04	Sprint 5	Planned
Should	US-REP-01	EP-07	As an administrator, I want student reports, so that academic records can be reviewed systematically.	Reporting	5	US-STU-04	Sprint 5	Planned
Should	US-REP-02	EP-07	As an administrator, I want attendance reports, so that trends and shortage cases can be reviewed.	Reporting	5	US-ATT-03	Sprint 5	Planned
Should	US-REP-03	EP-07	As a department	Decision support	8	US-EXM-05	Sprint 5	Planned

Priority	Story ID	Epic	User Story	Business value	SP	Dependency	Sprint	Status
			head, I want performance summaries, so that academic performance can be monitored.					
Could	US-REP-04	EP-07	As an administrator, I want course reports, so that enrollment and course information can be reviewed.	Reporting	5	US-CRS-04	Sprint 5	Planned

Prioritization Logic

1. Protect the foundation: authentication and role control precede protected academic operations.
2. Deliver a thin end-to-end academic path: student identity and course enrollment enable attendance and assessment workflows.
3. Prioritize stories that unlock other stories or reduce security/data-integrity risk.
4. Prefer user-visible outcomes over isolated technical work when ordering the product backlog.
5. Keep reporting and notifications behind the underlying source data so the team does not build reports on unstable information.
6. Defer optional features when they threaten the Sprint Goal or MVP stability.

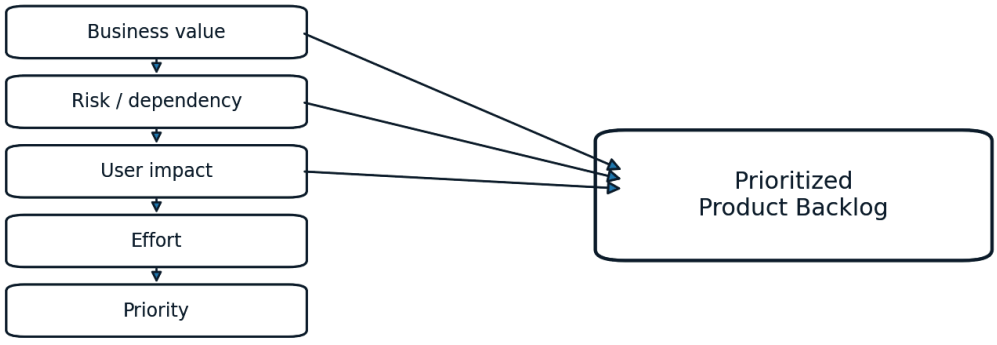


Figure 6. Project-specific backlog prioritization factors.

9. Story Point Estimation

Story points estimate relative effort and complexity rather than hours. The team uses a Fibonacci-style scale because uncertainty grows as work becomes more complex. Points should not be treated as individual performance scores.

Points	Interpretation	Typical example
1	Trivial	Logout or small UI adjustment
2	Small	Simple validation or display change
3	Moderate	Simple search/view workflow
5	Medium/high	CRUD workflow with validation and permissions
8	Complex	Attendance or marks workflow involving multiple rules
13	Very large / uncertain	A cross-cutting feature that should usually be split

Representative Estimation Reasoning

Story	Estimate	Reasoning
US-AUT-02 Login	3	Moderate UI + authentication flow, validation and session handling; dependencies are limited once registration exists.
US-STU-01 Add Student	5	Form validation, unique identifier rules, persistence and authorization.
US-ATT-01 Record Attendance	8	Course assignment, date/session rules, duplicate prevention, batch data and verification make it more complex.
US-EXM-03 Grade Calculation	8	Depends on validated marks and configurable academic rules; correctness matters more than UI complexity.
US-EXM-05 SGPA/CGPA	8	Requires aggregation across subjects/credits and careful rule handling; likely to expose clarification needs.

Planning Poker Example

Story under discussion: US-ATT-01 Record Attendance. Four team members independently reveal 5, 8, 8 and 13. The 13-point estimate highlights uncertainty around duplicate-session rules and the expected attendance-entry interface. After discussing the uncertainty, the team agrees that the core story is 8 points and that advanced bulk-import behavior should be split into a future story. The estimate therefore reflects the refined story, not the larger original concept.

10. Sprint Planning

The plan uses a short Foundation Sprint followed by five feature sprints. The exact capacity can be adjusted after the first sprint because velocity is unknown at the start of the project.

Sprint	Goal	Selected stories	Planned SP	Focus	Expected increment
Sprint 0	Foundation	Repository, project structure, environment, architecture, CI and basic data model	12	Engineering foundation	No feature claims; deployable skeleton
Sprint 1	Authentication + Student Foundation	US-AUT-01..05; US-STU-01..04	37	Secure identity and usable student records	Authenticated MVP foundation
Sprint 2	Course + Enrollment	US-CRS-01..04	16	Course catalogue and enrollment	Course participation workflow
Sprint 3	Attendance	US-ATT-01..04	21	Record, view and monitor attendance	Attendance increment
Sprint 4	Exams + Results	US-EXM-01..05	34	Marks, validation and results	Assessment/result increment
Sprint 5	Notifications + Reporting + Hardening	US-NOT-01..03; US-REP-01..04	34	Operational visibility, alerts and stabilization	Release candidate

Sprint 1 Planning Detail

Item	Details
Sprint Goal	Enable secure access and establish the student record foundation.
Selected stories	US-AUT-01, US-AUT-02, US-AUT-03, US-AUT-04, US-AUT-05, US-STU-01, US-STU-02, US-STU-03, US-STU-04
Dependencies	Authentication before protected student workflows; role assignment before admin-only actions.
Sprint backlog	UI forms; authentication API; session handling; RBAC checks; student CRUD; search; validation; automated tests; acceptance testing.
Expected increment	A user can authenticate, receive an appropriate role, and authorized users can create/search/update/view student records.
Definition of Done	Project-specific DoD from Section 7.



Figure 7. Sprint planning flow used to move from product intent to a committed sprint backlog.

11. Sprint Backlog

A Sprint Backlog converts selected stories into actionable work. The following example is for Sprint 3 because attendance provides a good demonstration of cross-functional work.

Story	Task	Owner area	Status	Task effort
US-ATT-01	Attendance entry screen	Frontend	To Do	3
US-ATT-01	Attendance API and persistence	Backend/DB	In Progress	3
US-ATT-01	Assigned-course authorization	Backend/Security	Code Review	2
US-ATT-01	Duplicate session validation	Backend/QA	To Do	2
US-ATT-02	Student attendance view	Frontend/API	Testing	3
US-ATT-03	Attendance report query	Backend/Reporting	To Do	3
US-ATT-04	Threshold calculation	Backend	To Do	2
US-ATT-04	Shortage notification trigger	Backend	To Do	2
ALL	Regression and acceptance testing	QA	In Progress	—

Sprint Backlog Rules

- Tasks support the Sprint Goal and selected stories; unrelated work is not added simply because a person has spare capacity.
- Technical tasks can move independently across the board while the team remains accountable for the story outcome.
- If a task reveals major uncertainty, the team discusses it during the Daily Scrum rather than silently carrying the risk.
- Stories are not considered Done merely because coding is finished; the Definition of Done still applies.

12. Scrum Board

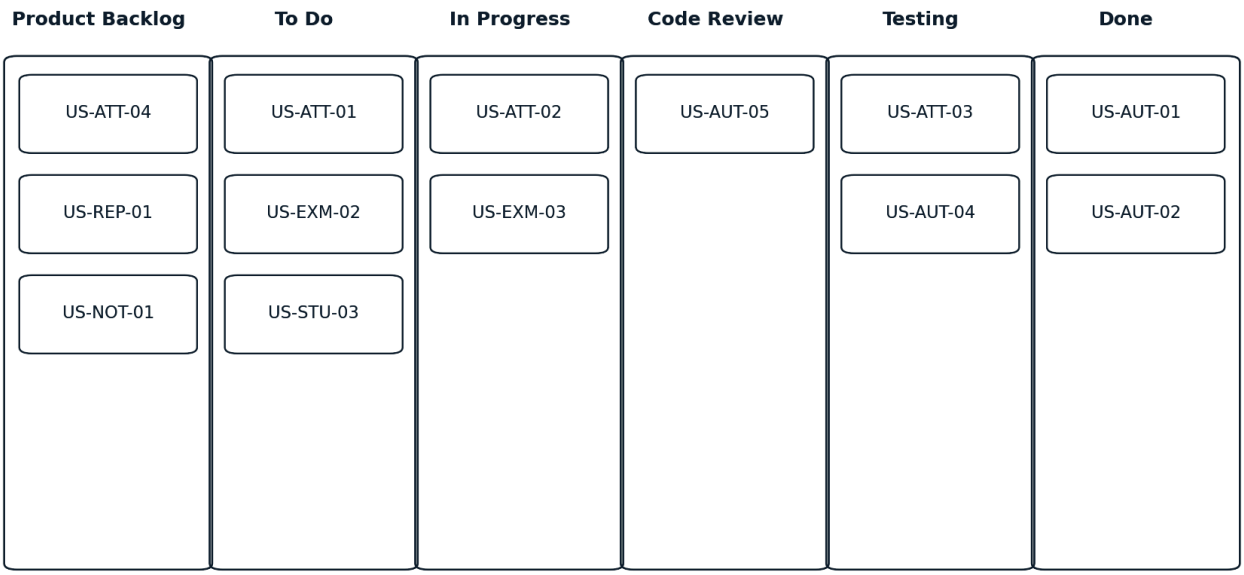


Figure 8. Example Scrum Board populated with realistic project stories.

The board makes workflow visible. A story normally moves from the product backlog into the sprint, then through To Do, In Progress, Code Review, Testing and Done. The exact board status should reflect the team's actual working policy.

How the Board Changes During a Sprint

- At Sprint Planning, selected stories are moved into the sprint's To Do area.
- As development starts, the active story moves to In Progress; limiting work-in-progress helps the team finish before starting too many new items.
- After implementation, review and testing expose defects or missing acceptance criteria. The card can move backward rather than being incorrectly marked Done.
- Done means the story satisfies the acceptance criteria and Definition of Done, not merely that the developer stopped working on it.

13. Burndown Chart

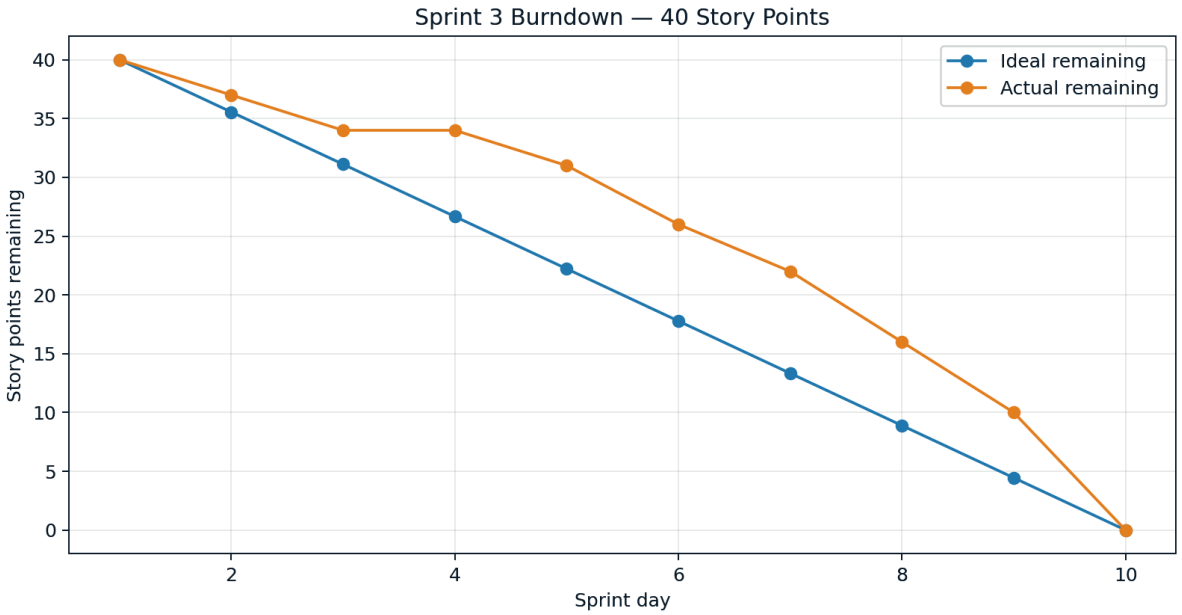
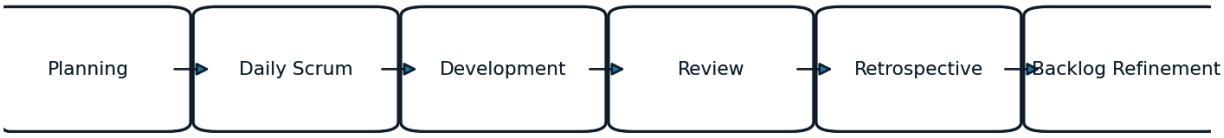


Figure 9. Sprint 3 burndown with 40 planned story points and a small mid-sprint delay.

The example starts with 40 story points. The actual line remains above the ideal line for several days because one attendance validation issue takes longer than expected. The team catches up later in the sprint. This is more informative than drawing a perfectly straight line because the purpose of the burndown is to make remaining work and emerging risk visible.

Day	Ideal remaining	Actual remaining	Interpretation
1	40	40	Sprint starts
2	36	37	Normal early progress
3	31	34	Slight lag
4	27	34	Validation issue delays completion
5	22	31	Team identifies impediment
6	18	26	Focused work reduces backlog
7	13	22	Progress improves
8	9	16	Testing clears several stories
9	4	10	Final acceptance work
10	0	0	Sprint target completed

14. Scrum Ceremonies



The cycle creates transparency, inspection and adaptation.

Figure 10. Ceremony cycle supporting transparency, inspection and adaptation.

Sprint Planning

Element	Project-specific approach
Purpose	Agree on a realistic Sprint Goal and select the work needed to achieve it.
Participants	Product Owner, Scrum Master and Development Team.
Inputs	Ordered backlog, product goal, known dependencies, capacity and previous learning.
Activities	Clarify stories, estimate/refine, select work, identify dependencies and formulate the Sprint Goal.
Outputs	Sprint Goal, Sprint Backlog and shared understanding of the expected increment.

Daily Scrum

The team uses the Daily Scrum to inspect progress toward the Sprint Goal and adapt the plan for the next day. The three practical questions used in this project are:

- What did I complete?
- What will I work on next?
- What impediments do I have?

Example: A developer reports that the attendance API is complete but duplicate-session validation is blocked by an unclear rule. The Product Owner clarifies the intended behavior after the meeting, while the team continues with test preparation. The impediment is visible and owned rather than hidden.

Sprint Review

The team demonstrates the working increment to relevant stakeholders. For this project, a review could demonstrate a student logging in, a faculty member recording attendance, a student viewing attendance, and an administrator inspecting a report. Stakeholder feedback is captured as backlog changes.

Example review outcome: stakeholders accept the attendance entry workflow but request that the report show course-wise percentages. The request becomes a new/refined backlog item rather than an unplanned task inserted into the completed sprint.

Sprint Retrospective

The retrospective focuses on the team's process, collaboration and quality—not on blaming individuals.

Issue	Impact	Root Cause	Action	Owner
Unclear grading rule	Rework in result story	Rule was assumed rather than confirmed	Create rule checklist before refinement	Product Owner
Too many items started at once	Testing queue increased	No WIP discipline	Limit active stories and finish before pulling new work	Scrum Master
Late UI feedback	Attendance screen needed rework	Review occurred too late	Show wireframe/prototype during refinement	Frontend + PO
Test data prepared late	Acceptance testing slowed	Data preparation not planned	Create reusable test dataset before sprint start	QA

15. Agile Framework Comparison

Framework	Core idea	Strengths	Weaknesses	Best suited for	Fit here
Scrum	Timeboxed sprints, roles, events and increments	Clear cadence; good for evolving requirements	Requires discipline and planned events	Product teams with changing priorities	Excellent
Kanban	Continuous flow and WIP limits	Flexible; strong visibility and flow	Less prescriptive about planning cadence	Support/operations or continuous-flow work	Very good
Extreme Programming (XP)	Engineering practices and rapid feedback	Strong code quality and testing practices	Requires mature engineering habits	Small software teams with quality focus	Good complement
Lean Development	Maximize value, reduce waste	Encourages efficiency and learning	Can be abstract for a student team without tailoring	Teams optimizing flow/value	Moderate
LeSS	Scale Scrum with minimal added structure	Keeps scaling relatively lean	Unnecessary complexity for one small product team	Multiple Scrum teams on one product	Poor fit
SAFe	Large-scale coordination across portfolios/programs	Useful for large enterprises	Heavy governance and planning overhead	Large organizations and portfolios	Poor fit
Scrum@Scale	Network of Scrum teams with scaling patterns	Supports coordination across many teams	Overkill for a single small team	Organizations with multiple Scrum teams	Poor fit

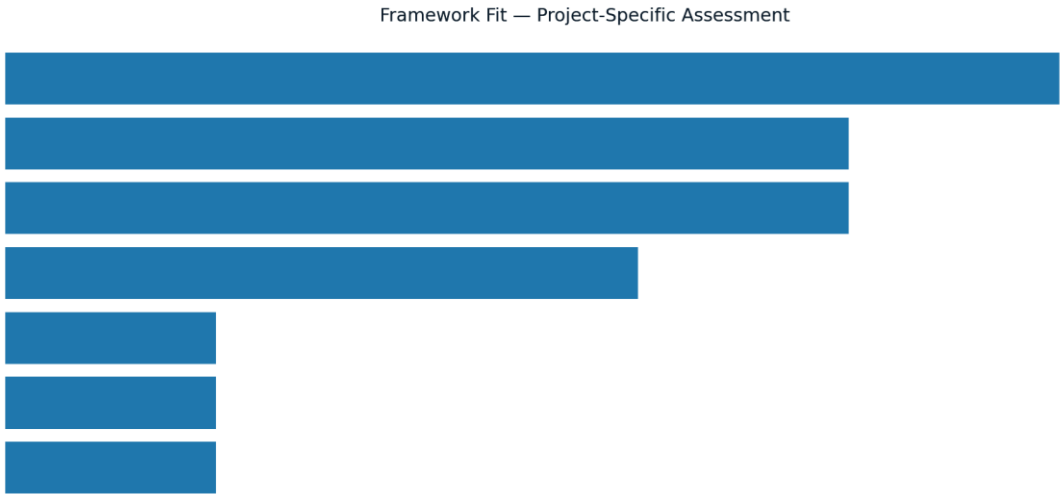


Figure 11. Project-specific fit assessment; this is a planning judgment, not a universal framework ranking.

Recommended Approach

For this contained B.Tech project, the recommended primary framework is standard Scrum, supported by selected Kanban-style visualization and XP-inspired engineering practices. Scrum provides a clear sprint cadence, Product Owner accountability, review/retrospective loops and a practical structure for the

assignment. A Kanban-style board improves flow visibility, while practices such as automated testing, code review and continuous integration strengthen technical quality.

LeSS, SAFe and Scrum@Scale are not recommended because the project does not contain the multiple teams, portfolio governance or organizational scaling problems those approaches are designed to address. Lean principles remain useful as a mindset—especially reducing rework and avoiding low-value features—but they do not need to replace the project's primary Scrum structure.

16. Agile Metrics

Metric	Meaning	Use	Responsible interpretation
Velocity	Story points completed per sprint	Supports capacity forecasting after several sprints	Do not use as an individual productivity score.
Sprint Burndown	Remaining work over sprint time	Shows whether work is trending toward the Sprint Goal	Investigate causes rather than manipulating the chart.
Sprint Goal Success	Whether the intended outcome was achieved	Measures outcome beyond raw point totals	A sprint can succeed even if minor scope changes occur.
Story Completion Rate	Completed vs planned stories	Shows planning reliability	Interpret alongside story size and quality.
Defect Rate	Defects associated with increments	Highlights quality trends	Avoid optimizing for low reporting by hiding defects.
Cycle Time	Time from work start to completion	Identifies workflow bottlenecks	Use trends, not isolated cases.
Lead Time	Time from request to delivery	Shows responsiveness to stakeholder needs	Can be affected by backlog waiting time.
Escaped Defects	Defects discovered after release	Indicates gaps in testing/acceptance	Use for improvement, not blame.

Metrics Principle

The team should use metrics to learn about the system of work. A higher velocity is not automatically better, and a lower defect count is not automatically evidence of higher quality if defects are simply being under-reported. The most useful metrics are trends that help the team identify bottlenecks, uncertainty and quality problems.

17. Project Timeline and Release Roadmap

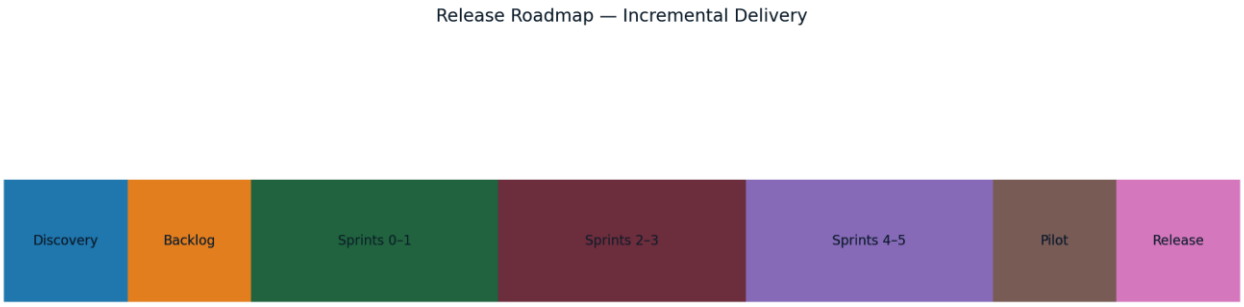


Figure 12. Incremental release roadmap from discovery through pilot and production release.

Phase	Purpose	Exit evidence
Discovery	Confirm problem, users, MVP boundaries and assumptions	Agreed product vision and initial backlog
Backlog preparation	Write/refine stories, acceptance criteria and estimates	Ordered backlog ready for sprint planning
Foundation / MVP	Build secure identity, student and course foundations	Usable academic foundation
Feature increments	Deliver attendance and examination/result workflows	Demonstrable increments with accepted stories
Operational hardening	Notifications, reporting, security, regression and performance checks	Release candidate meeting DoD
Pilot	Demonstrate representative workflows and gather feedback	Validated feedback and release decision
Production release	Package/deploy the agreed MVP scope	Release baseline and release notes
Future improvements	Refine based on real feedback and new priorities	New backlog items and release goals

18. Risks and Impediments

ID	Risk / impediment	Probability	Impact	Mitigation	Owner	Status
R-01	Unclear grading rules	Medium	High	Confirm rules during refinement; document configuration assumptions; add acceptance tests	Product Owner	Open
R-02	Changing requirements	High	Medium	Use backlog reprioritization and Sprint Goals instead of uncontrolled scope changes	PO + SM	Open
R-03	Poor test data quality	Medium	High	Prepare representative synthetic test data early	QA	Open
R-04	Authentication/security issue	Medium	High	RBAC tests, input validation, session checks and security review	Backend/QA	Open
R-05	Integration problems	Medium	Medium	Define API contracts early and integrate continuously	Backend + Frontend	Open
R-06	Sprint overcommitment	Medium	Medium	Use capacity-based planning and observe actual velocity after early sprints	Scrum Master	Open
R-07	Testing delay	Medium	High	Test continuously and avoid leaving all acceptance work to the last days	QA + Team	Open
R-08	Low user adoption	Low/Medium	Medium	Use review feedback and simple workflows; demonstrate value incrementally	PO	Open

Agile Risk Response

Agile does not remove project risk. It reduces the time between introducing a risky assumption and learning whether that assumption is correct. Short sprints, visible work, acceptance criteria and stakeholder reviews create repeated opportunities to detect and respond to risk.

19. Traceability



Figure 13. Traceability chain from business objective to release.

Business objective	Epic	User story	Acceptance criteria	Sprint	Release
Reduce manual academic record effort	EP-02 Student Management	US-STU-01 Add student	Valid mandatory fields; duplicate identifier rejected	Sprint 1	MVP / Release 1
Improve attendance visibility	EP-04 Attendance	US-ATT-02 View attendance	Authorized student sees current percentage	Sprint 3	Release 2
Improve assessment consistency	EP-05 Exams & Performance	US-EXM-03 Calculate grade	Configured rule produces consistent grade	Sprint 4	Release 2
Improve academic monitoring	EP-07 Reporting	US-REP-03 Performance summary	Authorized department view shows validated summary	Sprint 5	Release 3

Traceability is useful in Agile because it prevents backlog items from becoming disconnected feature lists. When a stakeholder asks for a change, the team can ask which business objective it supports, which epic/story should change, how acceptance criteria are affected, and which release should carry the value.

20. Complete Agile Documentation

Agile Artifacts Produced

Artifact	Purpose	Owner / participants
Product Vision	Defines desired product outcome and boundaries	Product Owner + team
Epic Map	Shows major value areas and hierarchy	Product Owner + team
User Story Repository	Represents user outcomes	Product Owner + team
Acceptance Criteria	Makes important stories testable	PO + Development/QA
Product Backlog	Orders work by value and dependency	Product Owner
Story Point Estimates	Creates relative effort/uncertainty understanding	Development Team
Sprint Backlog	Makes selected sprint work actionable	Development Team
Scrum Board	Makes workflow transparent	Entire team
Burndown	Shows remaining sprint work	Entire team
Review Notes	Captures stakeholder feedback and decisions	PO + team
Retrospective Actions	Tracks process improvements	Entire team
Risk Register	Makes uncertainty visible and actionable	PO + SM + team
Traceability Table	Connects value to delivery	PO + team
Release Roadmap	Communicates incremental delivery	PO + team

Final Agile Project Summary

The project plan converts the academic management platform into a sequence of manageable, testable increments. The highest-value foundation—identity, roles, student records and course participation—is established before attendance and examination workflows. This ordering reduces dependency risk and gives the team a usable product foundation early.

Scrum provides the primary delivery structure: a Product Owner orders the backlog, the Scrum Master supports the process, and the Development Team turns selected stories into increments. Acceptance criteria, a project-specific Definition of Done, the Scrum Board and burndown provide transparency. Sprint Reviews create a formal route for stakeholder feedback, while Retrospectives turn process problems into improvement actions.

The result is not a claim that the complete platform has already been implemented. It is an Agile planning and documentation baseline that can guide implementation, testing and future backlog refinement.

Final Submission Checklist

No.	Teacher-required deliverable	Location
1	Study the Business Problem and Product Vision	Covered in Sections 1–3
2	Agile Product Vision	Section 3
3	Scrum Team Structure	Section 4 + Figure 3
4	Scrum Roles and Responsibilities	Section 4 responsibility matrix
5	Project Epics	Section 5 + Figure 4
6	User Stories	Section 6 repository
7	Acceptance Criteria	Section 7
8	Prioritized Product Backlog	Section 8
9	Story Point Estimation	Section 9 + Planning Poker
10	Sprint Planning	Section 10
11	Sprint Backlog	Section 11
12	Scrum Board	Section 12 + Figure 8
13	Burndown Chart	Section 13 + Figure 9
14	Scrum Ceremonies	Section 14 + Figure 10
15	Sprint Review	Section 14
16	Sprint Retrospective	Section 14
17	Agile Framework Comparison	Section 15 + Figure 11
18	Recommended Agile Framework	Section 15
19	Project Timeline and Release Roadmap	Section 17 + Figure 12
20	Complete Agile Documentation	Section 20

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Assumptions Made

- The project is a contained B.Tech software-engineering mini-project, not a full institutional deployment.
- Specific institutional names, student populations and existing-system statistics were not invented.
- Academic grading and attendance thresholds are treated as configurable rules that require confirmation.
- Story points are illustrative relative estimates and should be recalibrated by the actual student team after Sprint 0.
- Sprint capacities are planning assumptions until actual velocity and team availability are observed.
- Notifications and reporting are scheduled after the core data workflows because they depend on stable source data.

Customization Before Submission

- Replace the generic project-team wording with actual team member names and roll numbers if required.
- Add the actual college/university name, department, academic year and guide name only if your submission format requires them.
- Adjust sprint dates to match your actual academic calendar.
- Replace illustrative story statuses with the team's real status if this document is used alongside implementation evidence.
- Confirm grading formulas, attendance thresholds and notification rules with the project requirements before implementation.

Closing Note

This Unit 2 document is designed to function as a genuine Agile project-planning artifact. Its emphasis is on decisions, dependencies, user value, measurable acceptance, iterative delivery and feedback rather than repeating general software-engineering theory.